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NOTES

Subject with code: Electronic Principles and Circuits (BEC303)

Module 3

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Syllabus :

Linear Opamp Circuits: Summing Amplifier and D/A Converter, Nonlinear Op-amp Circuits: Comparator with zero reference, Comparator with non-zero references. Comparator with Hysteresis.

Oscillator: Theory of Sinusoidal Oscillation, The Wein-Bridge Oscillator, RC Phase Shift Oscillator, The Colpitts Oscillator, Hartley Oscillator, Crystal Oscillator.

The 555 timer: Monostable Operation, Astable Operation.

Linear opamp circuits

Summing amplifier circuits

The subtracter

Figure 20-21 Subtractor.

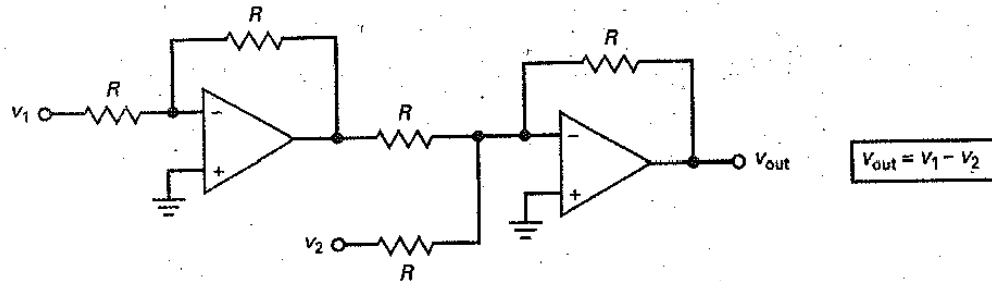


Figure 20.21 shows a circuit that subtract two input voltages to produce an output voltage equal to the difference of v_1 and v_2 .

Here how it works: input v_1 drives an inverter with a voltage gain of unity. The output of first stage amplifier is $-v_1$. This voltage is one of the inputs to the second stage summing circuit. The other input is v_2 . Since the gain of each channel is unity, the final output voltage is equals v_1 minus v_2 .

i.e $V_{out} = v_1 - v_2$

Summing on both inputs

Sometimes you may see a circuit like Fig. 20-22. It is nothing more than a summing circuit that has inverting and noninverting inputs. The inverting side of the amplifier has two input channels, and the noninverting side has two input channels. The total gain is the superposition of the channel gains.

The gain of each inverting channel is the ratio of the feedback resistor R_f to input channel resistance, either R_1 or R_2 . The gain of each noninverting channel is:

$$\frac{R_f}{R_1 \parallel R_2} + 1$$

reduced by the voltage-divider factor of the channel, either:

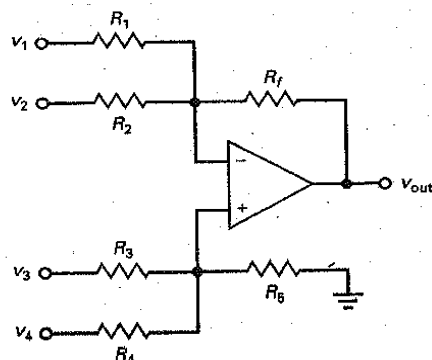
$$\frac{R_4 \parallel R_5}{R_3 + R_4 \parallel R_5}$$

or

$$\frac{R_3 \parallel R_5}{R_4 + R_3 \parallel R_5}$$

Figure 20-22 gives the equations for the gain of each channel. After getting each channel gain, we can calculate total output voltage.

Figure 20-22 Summing amplifier using both sides of op amp.



$$V_{out} = A_{v1}V_1 + A_{v2}V_2 + A_{v3}V_3 + A_{v4}V_4$$

$$A_{v1} = \frac{-R_f}{R_1}$$

$$A_{v2} = \frac{-R_f}{R_2}$$

$$A_{v3} = \left(\frac{R_f}{R_1 \parallel R_2} + 1 \right) \left(\frac{R_4 \parallel R_5}{R_3 + R_4 \parallel R_5} \right)$$

$$A_{v4} = \left(\frac{R_f}{R_1 \parallel R_2} + 1 \right) \left(\frac{R_3 \parallel R_5}{R_4 + R_3 \parallel R_5} \right)$$

Example 20-5

III Multisim

In Fig. 20-22, $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $R_3 = 3 \text{ k}\Omega$, $R_4 = 4 \text{ k}\Omega$, $R_5 = 5 \text{ k}\Omega$, and $R_f = 6 \text{ k}\Omega$. What is the voltage gain of each channel?

SOLUTION With the equations given in Fig. 20-22, the voltage gains are:

$$A_{v1} = \frac{-6 \text{ k}\Omega}{1 \text{ k}\Omega} = -6$$

$$A_{v2} = \frac{-6 \text{ k}\Omega}{2 \text{ k}\Omega} = -3$$

$$A_{v3} = \left(\frac{6 \text{ k}\Omega}{1 \text{ k}\Omega \parallel 2 \text{ k}\Omega} + 1 \right) \frac{4 \text{ k}\Omega \parallel 5 \text{ k}\Omega}{3 \text{ k}\Omega + 4 \text{ k}\Omega \parallel 5 \text{ k}\Omega} = 4.26$$

$$A_{v4} = \left(\frac{6 \text{ k}\Omega}{1 \text{ k}\Omega \parallel 2 \text{ k}\Omega} + 1 \right) \frac{3 \text{ k}\Omega \parallel 5 \text{ k}\Omega}{4 \text{ k}\Omega + 3 \text{ k}\Omega \parallel 5 \text{ k}\Omega} = 3.19$$

PRACTICE PROBLEM 20-5 Repeat Example 20-5 using $1 \text{ k}\Omega$ for R_f .

The Averager

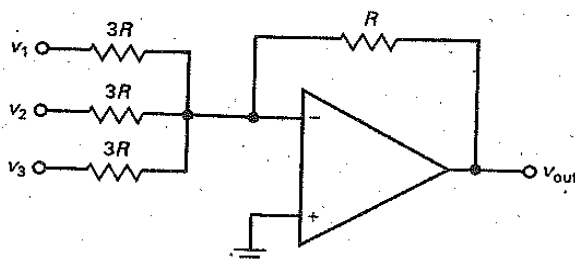
Figure 20-23 is an **averager**, a circuit whose output equals the average of the input voltages. Each channel has a voltage gain of:

$$A_v = \frac{R}{3R} = \frac{1}{3}$$

When all amplified outputs are added, we get an output that is the average of all input voltages.

The circuit shown in Fig. 20-23 has three inputs. Any number of inputs can be used, as long as each channel input resistance is changed to nR , where n is the number of channels.

Figure 20-23 Averaging circuit.



$$V_{out} = \frac{-(V_1 + V_2 + V_3)}{3}$$

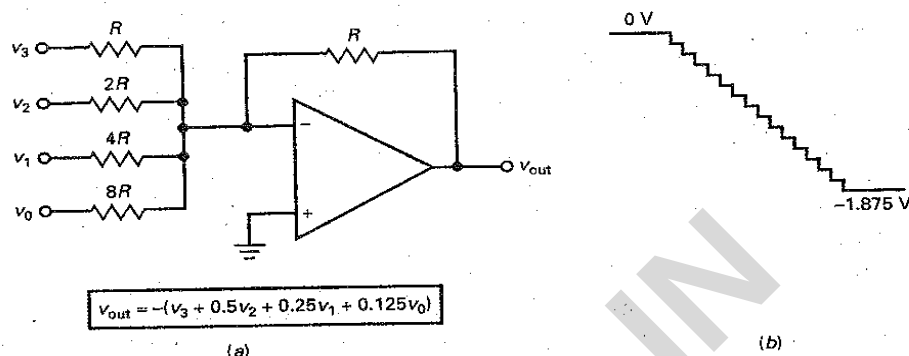
D/A Converter

In digital electronics, a digital-to-analog (D/A) converter takes a binary represented value and converts it into a voltage or current. This voltage or current will be proportional to the input binary value. There are two methods of D/A conversion often used.

1. The binary Weighted D/A converter
2. R/2R ladder D/A converter

The binary Weighted D/A converter

Figure 20-24 Binary-weighted D/A converter changes digital input to analog voltage.



The binary-weighted D/A converter is shown in Fig. 20-24. This circuit produces an output voltage equal to the weighted sum of the inputs. The *weight* is the same as the gain of the channel. For instance, in Fig. 20-24a the channel gains are:

$$A_{v3} = -1$$

$$A_{v2} = -0.5$$

$$A_{v1} = -0.25$$

$$A_{v0} = -0.125$$

The input voltages are digital or two-state, which means that they have a value of either 1 or 0. With 4 inputs, there are 16 possible input combinations of $v_3v_2v_1v_0$: 0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, and 1111.

When all inputs are zero (0000), the output is:

$$v_{out} = 0$$

When $v_3v_2v_1v_0$ is 0001, the output is:

$$v_{out} = -(0.125) = -0.125$$

When $v_3v_2v_1v_0$ is 0010, the output is:

$$v_{out} = -(0.25) = -0.25$$

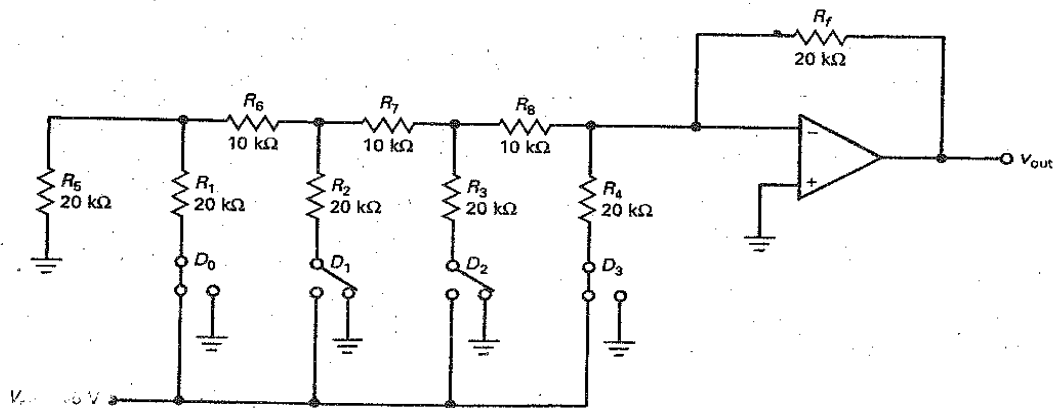
and so on. When the inputs are all 1s (1111), the output is maximum and equals:

$$v_{out} = -(1 + 0.5 + 0.25 + 0.125) = -1.875$$

When viewed on an oscilloscope the output voltage of the D/A converter will look like the negative going staircase shown in figure 20.24b.

R/2R ladder D/A converter

irc 20-25 R/2R ladder D/A converter.



The R/2R ladder D/A converter shown in figure 20.25 which is 4-bit D/A converter. The binary input switches D0-D3 connects 4 inputs to either ground (logic 0) or +Vref (logic 1). The ladder converts binary values from 0000 through 1111 to one of 16 unique output voltage levels. In the D/A converter shown in figure 20.25 D0 is least significant bit and D3 is most significant bit.

To determine the D/A converter's output voltage, you must first change the binary input value to its decimal equivalent value, BIN. This can be done by:

$$\text{BIN} = (D_0 \times 2^0) + (D_1 \times 2^1) + (D_2 \times 2^2) + (D_3 \times 2^3) \quad (20-18)$$

Then, the output voltage will be found by:

$$V_{\text{out}} = -\left(\frac{\text{BIN}}{2^N} \times 2V_{\text{ref}}\right) \quad (20-19)$$

where N equals the number of inputs.

Example 20-6

In Fig. 20-25, $D_0 = 1$, $D_1 = 0$, $D_2 = 0$, and $D_3 = 1$. Using a V_{ref} value of +5 V, determine the decimal equivalent of the binary input (BIN) and the output voltage of the converter.

SOLUTION Using Eq. 20-18, the decimal equivalent can be found by:

$$\text{BIN} = (1 \times 2^0) + (0 \times 2^1) + (0 \times 2^2) + (1 \times 2^3) = 9$$

The output voltage of the converter is found by using Eq. 20-19 as:

$$V_{\text{out}} = -\left(\frac{9}{2^4}\right) \times 2(5 \text{ V})$$

$$V_{\text{out}} = -\left(\frac{9}{16}\right)(10 \text{ V}) = -5.625 \text{ V}$$

Nonlinear Opamp Circuits

Comparators with zero reference

Non inverting Comparator

The simplest way to build a comparator is connect an op-amp without feedback resistor as shown in figure 22.1a. Due to high open loop gain +ve input produces +ve saturation and -ve input produce the -ve saturation.

The comparator of figure 22.1a is called zero crossing detector because the output voltage ideally switches from low to high and high to low whenever the input voltage crosses zero as

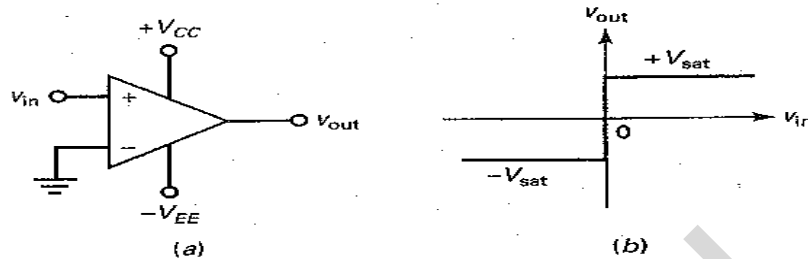
shown in figure 22.1b the input and output response. The minimum input voltage that produces saturation is.

$$v_{in(min)} = \frac{\pm V_{sat}}{A_{VOL}} \quad (22-1)$$

If $V_{sat} = 14$ V, the output swing of the comparator is from approximately -14 to $+14$ V. If the open-loop voltage gain is 100,000, the input voltage needed to produce saturation is:

$$v_{in(min)} = \frac{\pm 14 \text{ V}}{100,000} = \pm 0.14 \text{ mV}$$

Figure 22-1 (a) Comparator; (b) input/output response; (c) 741C response.

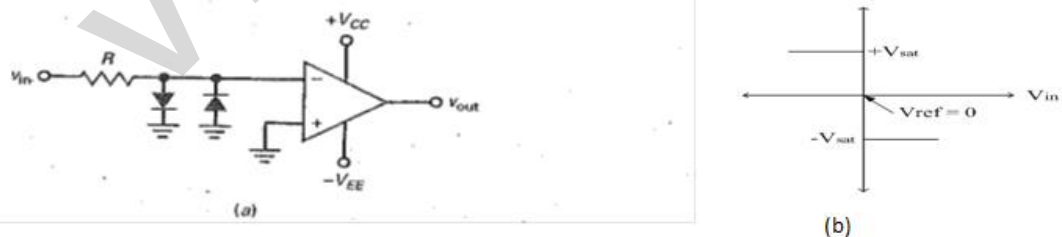


Inverting comparator

In inverting comparator the input signal to be compare is applied to the inverting terminal and non inverting terminal input is grounded shown in figure 22.2a. in this case, a slightly positive input voltage produces a maximum negative output as shown in figure 22.2b. On the other hand, a slightly negative input voltage produces a maximum positive output as shown in figure 22.2b.

The two diode clamps at the input terminal are used to protect comparator against excessively large input voltage. For some comparator maximum input voltage rating may be as little as $+5$ to -5 volts. Hence with these two diode clamps here we are maintaining input voltage $+0.7$ to -0.7 volts.

Figure 22-2 (a) Inverting comparator with clamping diodes; (b) input/output response.



Comparator with non zero references

In some applications a threshold voltage different from zero may be preferred. In figure 22.11a voltage divider produces the following reference voltage for the inverting input

$$v_{\text{ref}} = \frac{R_2}{R_1 + R_2} V_{CC} \quad (22-2)$$

When V_{in} is greater than V_{ref} , the differential input voltage is +ve and the output voltage is high. When V_{in} is less than V_{ref} , the differential input voltage is negative and the output voltage is low.

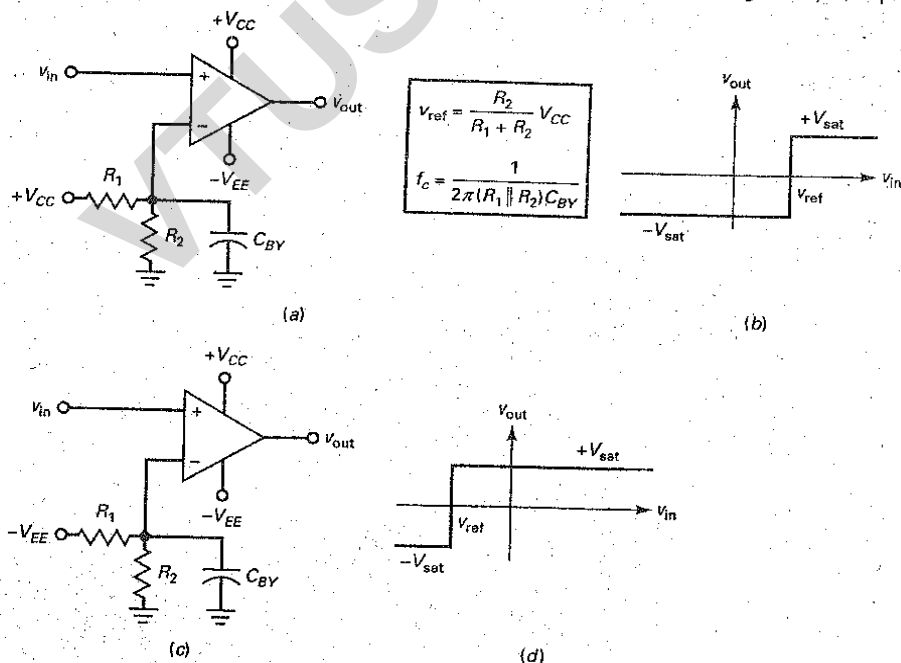
The bypass capacitor is typically used on the inverting input to reduce the amount of power supply ripple and other noise appearing at the inverting input. The cutoff frequency of this bypass circuit is

$$f_c = \frac{1}{2\pi(R_1 \parallel R_2)C_{BY}} \quad (22-3)$$

Figure 22.11b shows the transfer characteristic (input/output response). The trip point is now equal to V_{ref} . When V_{in} is greater than V_{ref} the output of comparator goes into +ve saturation. When V_{in} is less than V_{ref} the output goes -ve saturation.

If -ve reference is preferred connect $-V_{EE}$ to the voltage divider as shown in figure 22.11c. The Figure 22.11d shows the transfer characteristic (input/output response). The trip point is now equal to $-V_{ref}$. When V_{in} more +ve than the V_{ref} the output of comparator goes into +ve saturation. When V_{in} is more -ve than V_{ref} the output goes -ve saturation.

Figure 22-11 (a) Positive threshold; (b) positive input/output response; (c) negative threshold; (d) negative input/output response.



Comparators with Hyteresis

Schmitt Trigger

The standard solution for a noisy input is to use a comparator like the one shown in Fig. 22-18a. The input voltage is applied to the inverting input. Because the feedback voltage is aiding the input voltage, the feedback is *positive*. A comparator using positive feedback like this is usually called a **Schmitt trigger**.

When the comparator is positively saturated, a positive voltage is fed back to the noninverting input. This positive feedback voltage holds the output in the high state. Similarly, when the output voltage is negatively saturated, a negative

voltage is fed back to the noninverting input, holding the output in the low state. In either case, the positive feedback reinforces the existing output state.

The feedback fraction is:

$$B = \frac{R_1}{R_1 + R_2} \quad (22-4)$$

When the output is positively saturated, the reference voltage applied to the non-inverting input is:

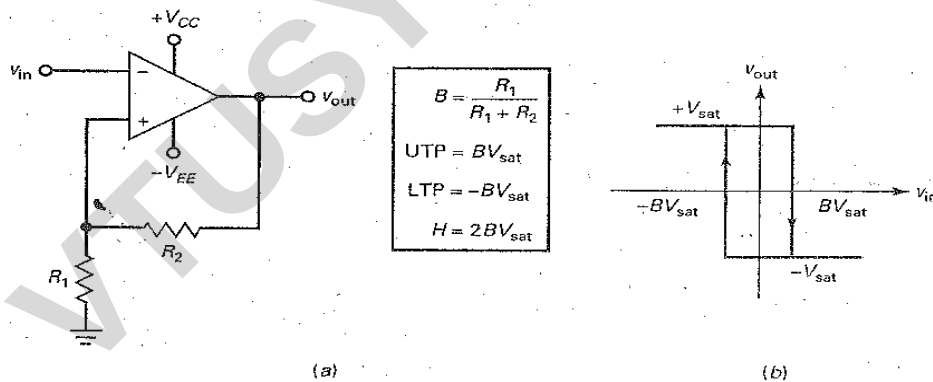
$$v_{\text{ref}} = +BV_{\text{sat}} \quad (22-5a)$$

When the output is negatively saturated, the reference voltage is:

$$v_{\text{ref}} = -BV_{\text{sat}} \quad (22-5b)$$

When the input voltage increased slightly more than $+BV_{\text{sat}}$ then output voltage switch from +ve to -ve as shown in figure 22.8b. Output voltage remains -ve untill the input voltage becomes more -ve than $-BV_{\text{sat}}$. Then the output switches from -ve to +ve.

Figure 22-18 (a) Inverting Schmitt trigger; (b) input/output response has hysteresis.



Hysteresis

The unusual response of Fig. 22-18b has a useful property called **hysteresis**. To understand this concept, put your finger on the upper end of the graph where it says $+V_{\text{sat}}$. Assume that this is the current value of output voltage. Move your finger to the right along the horizontal line. Along this horizontal line, the input voltage is changing but the output voltage is still equal to $+V_{\text{sat}}$. When you reach the upper right corner, v_{in} equals $+BV_{\text{sat}}$. When v_{in} increases to slightly more than $+BV_{\text{sat}}$, the output voltage goes into the transition region between the high and the low states.

If you move your finger down along the vertical line, you will simulate the transition of the output voltage from high to low. When your finger is on the lower horizontal line, the output voltage is negatively saturated and equal to $-V_{\text{sat}}$.

To switch back to the high output state, move your finger until it reaches the lower left corner. At this point, v_{in} equals $-BV_{\text{sat}}$. When v_{in} becomes slightly more negative than $-BV_{\text{sat}}$, the output voltage goes into the transition from low to high. If you move your finger up along the vertical line, you will simulate the switching of the output voltage from low to high.

In Fig. 22-18b, the trip points are defined as the two input voltages where the output voltage changes states. The *upper trip point (UTP)* has the value:

$$\text{UTP} = BV_{\text{sat}} \quad (22-6)$$

and the *lower trip point (LTP)* has the value:

$$\text{LTP} = -BV_{\text{sat}} \quad (22-7)$$

The difference between these trip points is defined as the hysteresis (also called the *deadband*):

$$H = \text{UTP} - \text{LTP} \quad (22-8)$$

With Eqs. (22-6) and (22-7), this becomes:

$$H = BV_{\text{sat}} - (-BV_{\text{sat}})$$

which equals:

$$H = 2BV_{\text{sat}} \quad (22-9)$$

Oscillator

Theory of Oscillations

To build a sinusoidal oscillator, we need amplifier with +ve feedback. The idea is to use the feedback signal in place of the input signal. If the feedback is large with correct phase, there will be output oscillations without external input signal.

Figure 23-1a shows an ac voltage source driving the input terminals of an amplifier. The amplified output voltage is:

$$v_{out} = A_v(v_{in})$$

This voltage drives a feedback circuit that is usually a resonant circuit. Because of this, we get maximum feedback at one frequency. In Fig. 23-1a, the feedback voltage returning to point x is given by:

$$v_f = A_v B(v_{in})$$

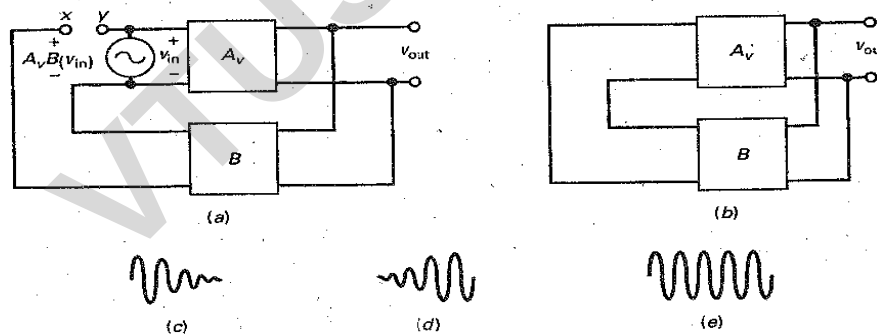
where B is the feedback fraction.

If the phase shift through the amplifier and feedback circuit is equivalent to 0° , $A_v B(v_{in})$ is in phase with v_{in} .

Suppose we connect point x to point y and simultaneously remove voltage source v_{in} . Then the feedback voltage $A_v B(v_{in})$ drives the input of the amplifier, as shown in Fig. 23-1b.

In any oscillator the loop gain $A_v B$ is greater than 1 when the power is first turned on. A small starting voltage is applied to the input terminals, and the output voltage builds up, as shown in Fig. 23-1d. After the output voltage reaches a certain level, $A_v B$ automatically decreases to 1, and the peak-to-peak output becomes constant (Fig. 23-1e).

Figure 23-1 (a) Feedback voltage returns to point x; (b) connecting points x and y; (c) oscillations die out; (d) oscillations increase; (e) oscillations are fixed in amplitude.



The Wien Bridge Oscillator

Lead-Lag Circuit

The Wien-bridge oscillator uses a resonant feedback circuit called a **lead-lag circuit** (Fig. 23-4). At very low frequencies, the series capacitor appears open to the input signal, and there is no output signal. At very high frequencies, the shunt capacitor looks shorted, and there is no output. In between these extremes, the output voltage reaches a maximum value (see Fig. 23-5a). The frequency where the output is maximum is the **resonant frequency f_r** . At this frequency, the feedback fraction B reaches a maximum value of $\frac{1}{3}$.

Formula for Resonant Frequency

By analyzing Fig. 23-4 with complex numbers, we can derive these two equations:

$$B = \frac{1}{\sqrt{9 - (X_C/R - R/X_C)^2}} \quad (23-1)$$

Figure 23-4 Lead-lag circuit.

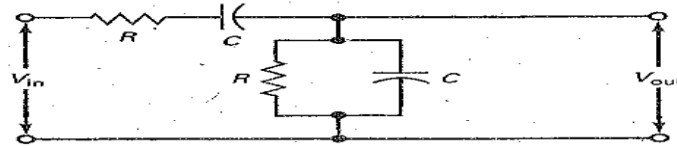
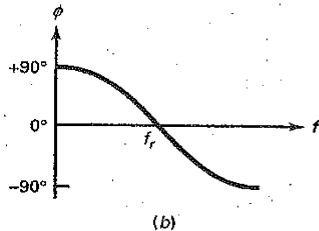
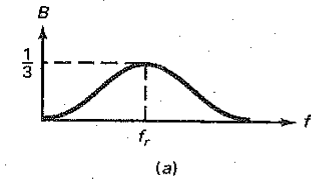


Figure 23-5 (a) Voltage gain; (b) phase response; (c) phasor diagram.



$$\phi = \arctan \frac{X_C/R - R/X_C}{3} \quad (23-2)$$

Graphing these equations produces Fig. 23-5a and b.

The feedback fraction given by Eq. (23-1) has a maximum value at the resonant frequency. At this frequency, $X_C = R$:

$$\frac{1}{2\pi f_r C} = R$$

Solving for f_r gives:

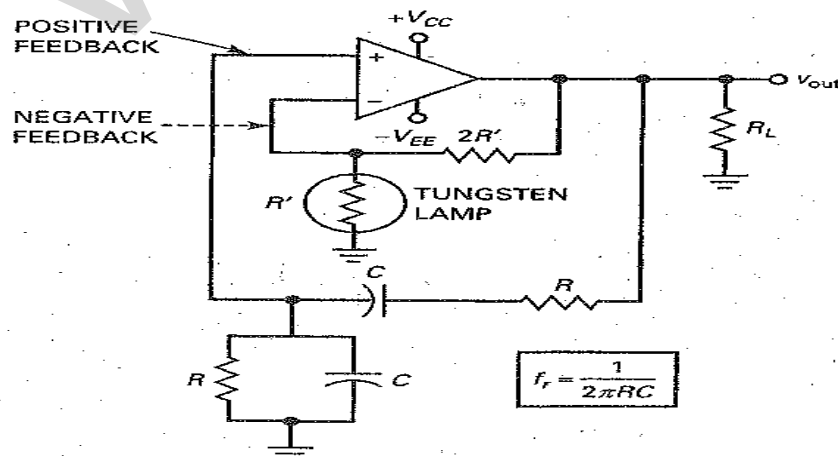
$$f_r = \frac{1}{2\pi RC} \quad (23-3)$$

Working of Wien Bridge Oscillator

Figure 23-6a shows a Wien-bridge oscillator. It uses positive and negative feedback because there are two paths for feedback. There is a path for positive feedback from the output through the lead-lag circuit to the noninverting input. There is also a path for negative feedback from the output through the voltage divider to the inverting input.

At some high output level, the tungsten lamp has a resistance of exactly R' . At this point, the closed-loop voltage gain from the noninverting input to the

Figure 23-6 Wien-bridge oscillator.



output decreases to:

$$A_{v(CL)} = \frac{2R'}{R'} + 1 = 3.$$

Since the lead-lag circuit has a B of $\frac{1}{3}$, the loop gain is:

$$A_{v(CL)}B = 3(\frac{1}{3}) = 1$$

When the power is first turned on, the resistance of the tungsten lamp is less than R' . As a result, the closed-loop voltage gain from the noninverting input to the output is greater than 3 and $A_{v(CL)}B$ is greater than 1.

As the oscillations build up, the peak-to-peak output becomes large enough to increase the resistance of the tungsten lamp. When its resistance equals R' , the loop gain $A_{v(CL)}B$ is exactly equal to 1. At this point, the oscillations become stable, and the output voltage has a constant peak-to-peak value.

RC Phase Shift Oscillator

Figure 23-13 is a phase-shift oscillator with three lead circuits in the feedback path. As you recall, a lead circuit produces a phase shift between 0° and 90° , depending on the frequency. At some frequency, the total phase shift of the three lead circuits equals 180° (approximately 60° each). Some phase-shift oscillator configurations use four lead circuits to produce the required 180° of phase shift. The amplifier has an additional 180° of phase shift because the signal drives the inverting input. As a result, the phase shift around the loop will be 360° , equivalent to 0° . If A_vB is greater than 1 at this particular frequency, oscillations can start.

Figure 23-14 shows an alternative design. It uses three lag circuits. The operation is similar. The amplifier produces 180° of phase shift, and the lag circuits contribute -180° at some higher frequency to get a loop phase shift of 0° . If A_vB is greater than 1 at this frequency, oscillations can start. The phase-shift oscillator is not a popular circuit. Again, the main problem with the circuit is that it cannot be easily adjusted over a large frequency range.

Figure 23-13 Phase-shift oscillator with three lead circuits.

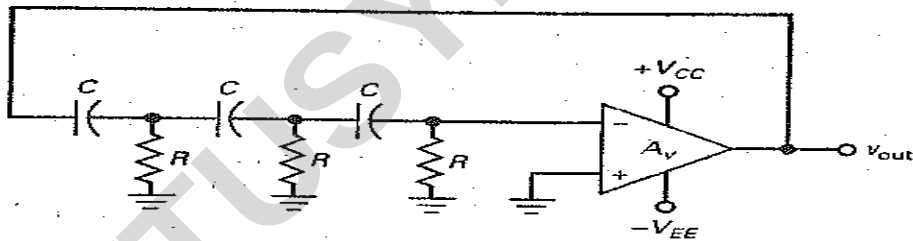
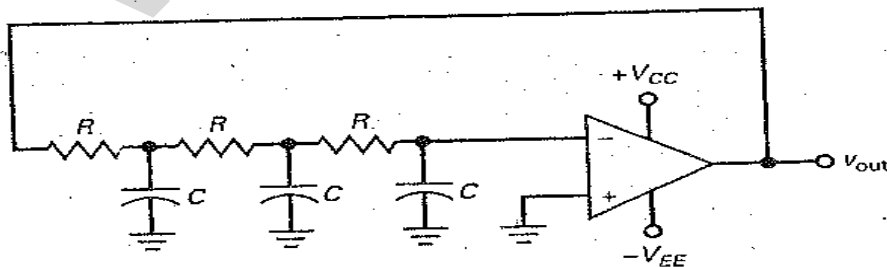


Figure 23-14 Phase-shift oscillator with three lag circuits.



The Colpitts Oscillator

Figure 23-15 shows a **Colpitts oscillator**. The voltage-divider bias sets up a quiescent operating point. The RF choke has a very high inductive reactance, so it appears open to the ac signal. The circuit has a low-frequency voltage gain of r_c/r'_e , where r_c is the ac collector resistance. Because the RF choke appears open to the ac signal, the ac collector resistance is primarily the ac resistance of the resonant tank circuit. This ac resistance has a maximum value at resonance.

Figure 23-15 Colpitts oscillator.

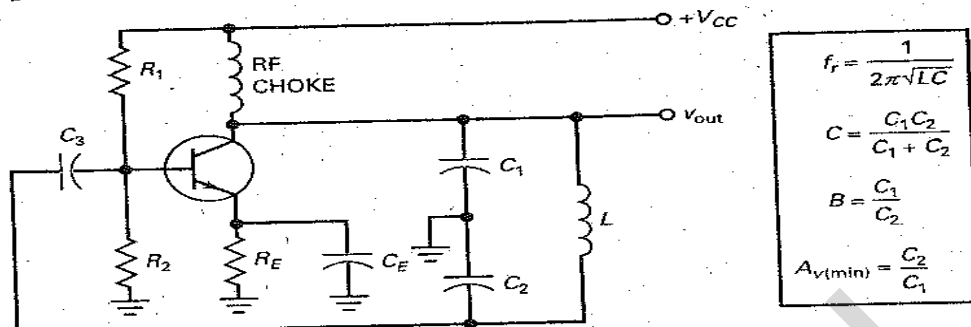
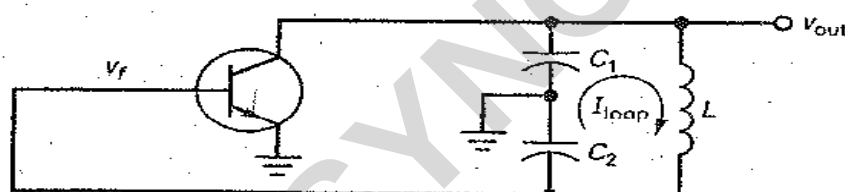


Figure 23-16 Equivalent circuit of Colpitts oscillator.



You will encounter many variations of the Colpitts oscillator. One way to recognize a Colpitts oscillator is by the capacitive voltage divider formed by C_1 and C_2 . It produces the feedback voltage necessary for oscillations.

Hartley Oscillator

Figure 23-22 Hartley oscillator.

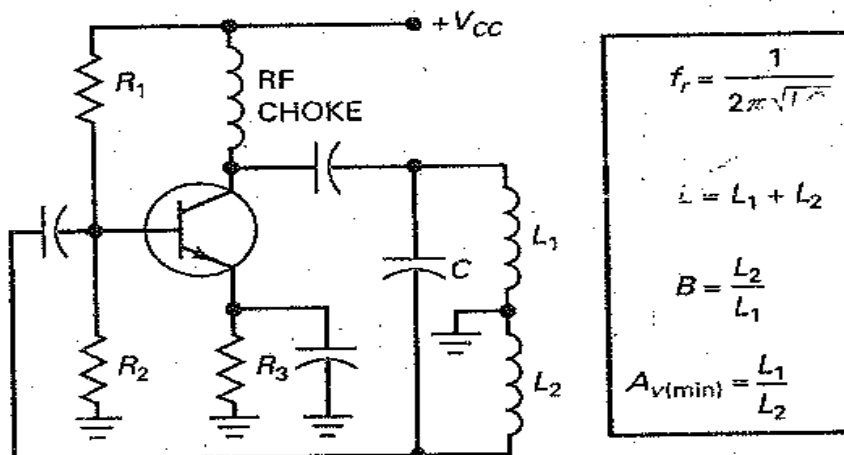


Figure 23-22 is an example of the **Hartley oscillator**. When the LC tank is resonant, the circulating current flows through L_1 in series with L_2 . The equivalent L to use in Eq. (23-5) is:

$$L = L_1 + L_2 \quad (23-15)$$

In a Hartley oscillator, the feedback voltage is developed by the inductive voltage divider, L_1 and L_2 . Since the output voltage appears across L_1 and the feedback voltage appears across L_2 , the feedback fraction is:

$$B = \frac{L_2}{L_1} \quad (23-16)$$

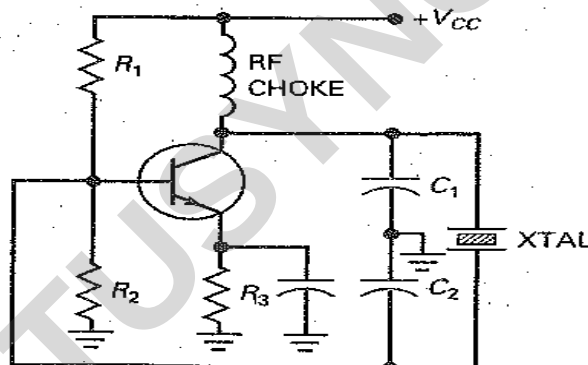
As usual, this ignores the loading effects of the base. For oscillations to start, the voltage gain must be greater than $1/B$.

Often a Hartley oscillator uses a single tapped inductor instead of two separate inductors. Another variation sends the feedback signal to the emitter instead of to the base. Also, you may see an FET used instead of a bipolar junction transistor. The output signal can be either capacitively coupled or link-coupled.

Crystal Oscillator

When accuracy and stability of the oscillation frequency are important, a **quartz-crystal oscillator** is used. In Fig. 23-24, the feedback signal comes from a

Figure 23-24 Crystal oscillator.



capacitive tap. As will be discussed in the next section, the crystal (abbreviated *XTAL*) acts like a large inductor in series with a small capacitor (similar to the Clapp). Because of this, the resonant frequency is almost totally unaffected by transistor and stray capacitances.

The 555 Timer

The NE555 (also LM555, CA555, and MC1455) is a widely used *IC timer*, a circuit that can run in either of two modes: **monostable** (one stable state) or **astable** (no stable states). In the monostable mode, it can produce accurate time delays from microseconds to hours. In the astable mode, it can produce rectangular waves with a variable duty cycle.

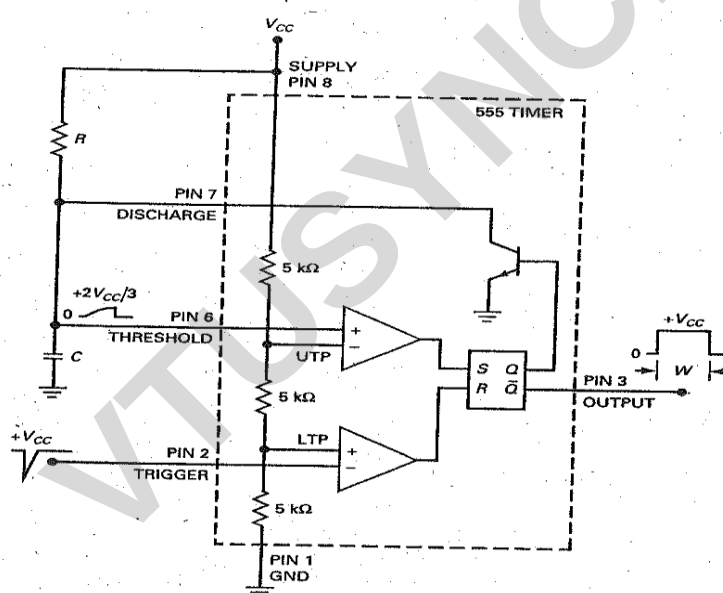
Monostable Operation

Figure 23-33 shows the 555 timer connected for monostable operation. The circuit has an external resistor R and a capacitor C . The voltage across the capacitor is used for the threshold voltage to pin 6. When the trigger arrives at pin 2, the circuit produces a rectangular output pulse from pin 3.

Here is the theory of operation. Initially, the Q output of the RS flip-flop is high. This saturates the transistor and clamps the capacitor voltage at ground. The circuit will remain in this state until a trigger arrives. Because of the voltage divider, the trip points are the same as previously discussed: $UTP = 2V_{CC}/3$ and $LTP = V_{CC}/3$.

When the trigger input falls to slightly less than $V_{CC}/3$, the lower comparator resets the flip-flop. Since Q has changed to low, the transistor goes into cutoff, allowing the capacitor to charge. At this time, $\bar{Q}$ has changed to high. The

Figure 23-33 555 timer connected for monostable operation.



capacitor now charges exponentially as shown. When the capacitor voltage is slightly greater than $2V_{CC}/3$, the upper comparator sets the flip-flop. The high Q turns on the transistor, which discharges the capacitor almost instantly. At the same instant, $\bar{Q}$ returns to the low state and the output pulse ends. $\bar{Q}$ remains low until another input trigger arrives.

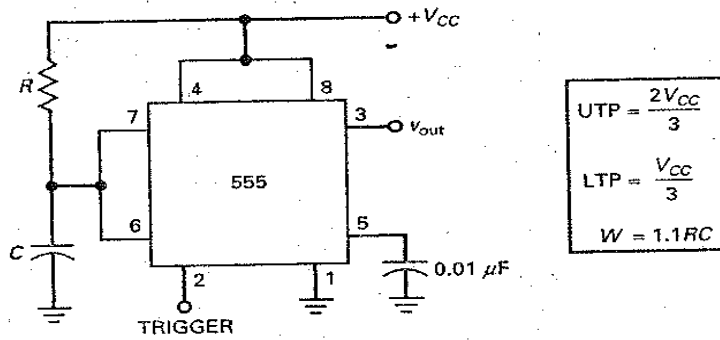
The formula for pulse width generated by monostable

$$W = 1.1RC$$

(23-25)

Figure 23-34 shows the schematic diagram for the monostable 555 circuit as it usually appears. Only the pins and external components are shown.

Figure 23-34 Monostable timer circuit.



Astable Operation

Figure 23-35 shows the 555 timer connected for astable operation. The trip points are the same as for monostable operation:

$$UTP = \frac{2V_{CC}}{3}$$

$$LTP = \frac{V_{CC}}{3}$$

When Q is low, the transistor is cut off and the capacitor is charging through a total resistance of:

$$R = R_1 + R_2$$

Because of this, the charging time constant is $(R_1 + R_2)C$. As the capacitor charges, the threshold voltage (pin 6) increases.

Eventually, the threshold voltage exceeds $+2V_{CC}/3$. Then, the upper comparator sets the flip-flop. With Q high, the transistor saturates and grounds pin 7. The capacitor now discharges through R_2 . Therefore, the discharging time constant is R_2C . When the capacitor voltage drops to slightly less than $V_{CC}/3$, the lower comparator resets the flip-flop.

Figure 23-36 shows the waveforms. The timing capacitor has exponentially rising and falling voltages between UTP and LTP. The output is a rectangular wave that swings between 0 and V_{CC} . Since the charging time constant is longer than the discharging time constant, the output is nonsymmetrical. Depending on resistances R_1 and R_2 , the duty cycle is between 50 and 100 percent.

By analyzing the equations for charging and discharging, we can derive the following formulas. The pulse width is given by:

$$W = 0.693(R_1 + R_2)C \quad (23-26)$$

The figure 23.37 shows the schematic diagram for astable 555 circuit.

Figure 23-35 555 timer connected for astable operation.

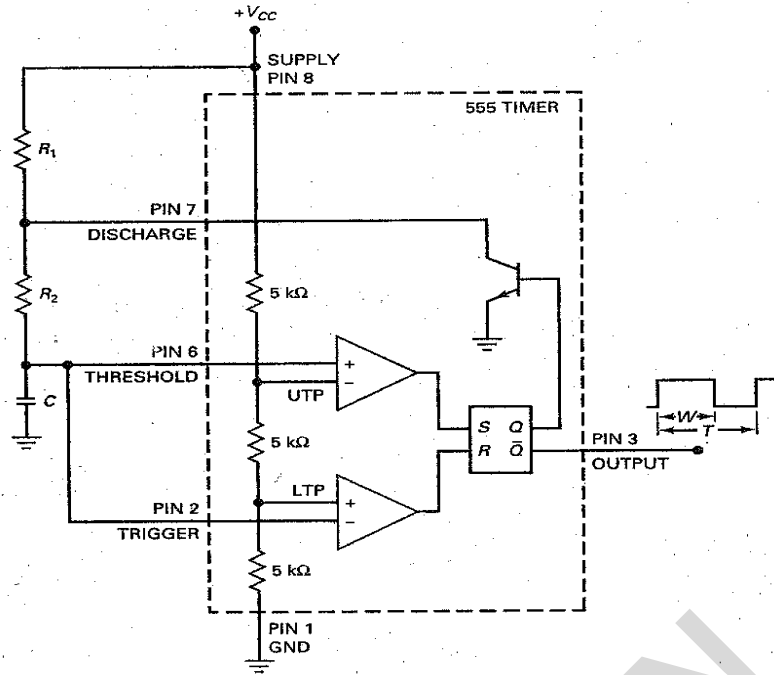
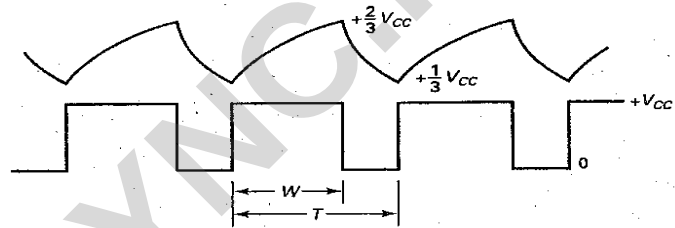


Figure 23-36 Capacitor and output waveforms for astable operation.



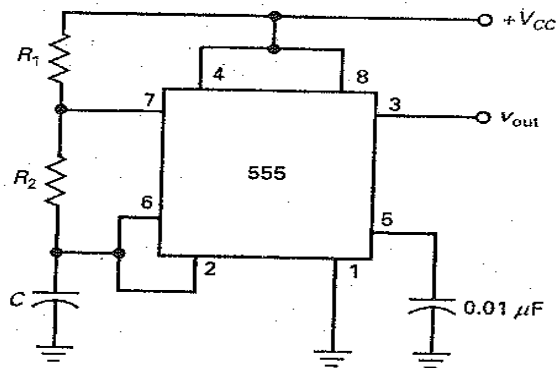
The period of the output equals:

$$T = 0.693(R_1 + 2R_2)C \quad (23-27)$$

The reciprocal of the period is the frequency:

$$f = \frac{1.44}{(R_1 + 2R_2)C} \quad (23-28)$$

Figure 23-37 Astable multivibrator.



$$\begin{aligned} W &= 0.693(R_1 + R_2)C \\ T &= 0.693(R_1 + 2R_2)C \\ f &= \frac{1.44}{(R_1 + 2R_2)C} \\ D &= \frac{R_1 + R_2}{R_1 + 2R_2} \end{aligned}$$

Dividing the pulse width by the period gives the duty cycle:

$$D = \frac{R_1 + R_2}{R_1 + 2R_2} \quad (23-29)$$